

Publication List

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Published Papers(* donotes equal contribution)

- [1] W. Cao, T. Zhang, Z. Sun, C. Liu, S. S.-T. Yau, and S. E. Li. “Nonlinear Bayesian Filtering With Natural Gradient Gaussian Approximation”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 48.7 (2026), pp. 8581–8597.
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- [3] K. Zhou, Z. Wen, Z. Zhuo, Z. Yan, P. Wu, S. Li, H. Gao, K. Ding, W. Cao, W. Pan, and C. Liu. “CoINS: Counterfactual Interactive Navigation via Skill-Aware VLM”. In: *IEEE Transactions on Automation Science and Engineering* (2026). arXiv preprint arXiv:2601.03956.
- [4] T. Zhang, W. Cao, and S. E. Li. “Natural Gradient Gaussian Approximation Filter with Positive Definiteness Guarantee”. In: *2026 American Control Conference (ACC)*. 2026, pp. 3224–3229.
- [5] S. Liu*, W. Cao*, C. Liu, Z. He, T. Zhang, Y. Wang, and S. E. Li. “One Filters All: A Generalist Filter For State Estimation”. In: *Advances in Neural Information Processing Systems*. Vol. 38, Main Conference. Curran Associates, Inc., 2025, pp. 21276–21302.
- [6] T. Zhang, W. Cao, C. Liu, T. Zhang, J. Li, and S. E. Li. “Robust State Estimation for Legged Robots With Dual Beta Kalman Filter”. In: *IEEE Robotics and Automation Letters* 10.8 (2025), pp. 7955–7962.
- [7] W. Cao, C. Liu, Z. Lan, S. E. Li, W. Pan, and A. Alessandri. “Robust Bayesian Inference for Moving Horizon Estimation”. In: *Automatica* 173 (2025), p. 112108. ISSN: 0005-1098.
- [8] T. Zhang, W. Cao, C. Liu, F. Zhang, W. Wu, and S. E. Li. “NANO-SLAM: Natural Gradient Gaussian Approximation for Vehicle SLAM”. In: *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*. 2025, pp. 4014–4019.
- [9] W. Cao, A. Capone, R. Yadav, S. Hirche, and W. Pan. “Computation-Aware Learning for Stable Control with Gaussian Process”. In: *Proceedings of Robotics: Science and Systems*. Delft, Netherlands, July 2024.
- [10] W. Cao and W. Pan. “Impact of Computation in Integral Reinforcement Learning for Continuous-Time Control”. In: *The Twelfth International Conference on Learning Representations*. 2024.
- [11] S. Liu, W. Cao, C. Liu, T. Zhang, and S. E. Li. “Convolutional Unscented Kalman Filter for Multi-Object Tracking with Outliers”. In: *IEEE Transactions on Intelligent Vehicles* (2024), pp. 1–12.
- [12] J. Duan, W. Cao, Y. Zheng, and L. Zhao. “On the Optimization Landscape of Dynamic Output Feedback Linear Quadratic Control”. In: *IEEE Transactions on Automatic Control* 69.2 (2024), pp. 920–935.
- [13] W. Cao, C. Liu, Z. Lan, Y. Piao, and S. E. Li. “Generalized Moving Horizon Estimation for Nonlinear Systems with Robustness to Measurement Outliers”. In: *2023 American Control Conference*. 2023, pp. 1614–1621.
- [14] W. Wu, C. Liu, W. Cao, C. Chen, H. Li, and S. E. Li. “Multi-Sensor Calibration System for Mobile Robots”. In: *2023 7th CAA International Conference on Vehicular Control and Intelligence*. 2023, pp. 1–7.
- [15] W. Wang, Y. Zhang, J. Gao, Y. Jiang, Y. Yang, Z. Zheng, W. Zou, J. Li, C. Zhang, W. Cao, G. Xie, J. Duan, and S. E. Li. “GOPS: A General Optimal Control Problem Solver for Autonomous Driving and Industrial Control Applications”. In: *Communications in Transportation Research* 3 (2023), p. 100096.

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- [19] Z. Gu, Y. Yang, J. Duan, S. E. Li, J. Chen, W. Cao, and S. Zheng. “Belief State Separated Reinforcement Learning for Autonomous Vehicle Decision Making under Uncertainty”. In: *2021 IEEE International Intelligent Transportation Systems Conference*. 2021, pp. 586–592.
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- [23] X. Chen, K. Yan, W. Cao, and L. Zhao. “Taming OOD Actions for Offline Reinforcement Learning: An Advantage-Based Approach”. In: *arXiv preprint arXiv:2505.05126* (2025).